

$$dV_m = \sqrt{\det(g)} d^k u$$

$$= \sqrt{\det(J^T J)} d^k u$$

1-维流形, $\gamma(t)$

$$g = \sqrt{\gamma'(t)^2}$$

$$= |\gamma'(t)|$$

$$dV_m$$

$$= ds = \gamma'(t) dt$$

2-维流形
 $\Phi(u, v)$

$$J = \begin{bmatrix} \frac{1}{T_u} & \frac{1}{T_v} \\ 1 & 1 \end{bmatrix}$$

$$J^T = \begin{bmatrix} -T_u & - \\ -T_v & - \end{bmatrix}$$

$$J^T J = \begin{bmatrix} T_u \cdot T_u & T_u \cdot T_v \\ T_v \cdot T_u & T_v \cdot T_v \end{bmatrix} = g$$

$$g = \begin{bmatrix} T_u \cdot T_u & T_u \cdot T_v \\ T_v \cdot T_u & T_v \cdot T_v \end{bmatrix}$$

$$\det g = |T_u|^2 |T_v|^2 - (T_u \cdot T_v)^2 \\ = |T_u \times T_v|^2 \quad \text{三 维}$$

$$\int_C f(x, y, z) ds = \int_C f(x, y, z) \gamma'(t) dt \quad \text{~~~~~ } \rho(x, y, z)$$

$$\int_S f(x, y, z) \sqrt{\det(g)} du dv$$



$$\det \left(\begin{matrix} \vdots & \vdots \\ \vdots & \vdots \end{matrix} \right)^T \geq g_1 \cdots g_n$$

$$\int_M dw = \int_{\partial M} w$$

$$1. \int_a^b \frac{df}{dx} dx = f(b) - f(a)$$

平面格林公式

$$2. \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = \oint_{\partial D} (P dx + Q dy)$$

$$F = \begin{bmatrix} P(x,y) \\ Q(x,y) \end{bmatrix}$$

连续可微

二维旋度

3.

Surface

$$\iint_S (\nabla \times F) \cdot n dA = \oint_{\partial S} F \cdot dr$$

$$F = \begin{bmatrix} P(x,y,z) \\ Q(x,y,z) \\ R(x,y,z) \end{bmatrix}$$

斯托克斯公式

$$dr = \begin{bmatrix} dx \\ dy \\ dz \end{bmatrix}$$



$$\det \begin{bmatrix} \frac{\partial}{\partial x} & P \\ \frac{\partial}{\partial y} & Q \end{bmatrix} = \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$$

$$= \oint_{\partial D} F \cdot dr$$

有界光滑



$$4. \iiint_V (\nabla \cdot \mathbf{F}) dV = \iint_{\partial V} \mathbf{F} \cdot \mathbf{n} dA$$

V - 有界立体区域 ∂V - 分段光滑封闭边界

$$\mathbf{F} = \begin{bmatrix} P \\ Q \\ R \end{bmatrix}$$



$$1. \int_a^b \frac{df}{dx} dx = f(b) - f(a)$$

左边区域
[a, b]

左积分对象
 $\frac{df}{dx}$ 导数

右区域
 $\{b\} - \{a\}$
0 维

右积分对象
f

$$2. \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = \oint_{\partial D} (P dx + Q dy)$$

二维曲面

旋度

1 维曲线
 ∂S

F 环流

$$3. \iiint_S (\nabla \times F) \cdot n dA = \oint_{\partial S} F \cdot dr$$

三维体积

散度

2 维曲面
 ∂V

F 通量

$$4. \iiint_V (\nabla \cdot F) dv = \iint_{\partial V} F \cdot n dA$$

M-k 维
流形

$$\int_M dw = \int_{\partial M} w$$

k-1 维上

$$1\text{-form}$$

$$V = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

$V \rightarrow \frac{dx}{\downarrow}$ 向量 $\rightarrow$ 标量
1-form
线性映射

$$\gamma(t) = \begin{bmatrix} x(t) \\ y(t) \\ z(t) \end{bmatrix}$$

$$\gamma'(t) = \begin{bmatrix} x'(t) \\ y'(t) \\ z'(t) \end{bmatrix}$$

$$dx(v) = v_1$$

$$dy(v) = v_2$$

$$dz(v) = v_3$$

微分几何

$$dx(\gamma'(t)) = x'(t) = \dot{x}(t)$$

莱布尼兹微积分

$$dx = x'(t) dt$$

dx — 无穷小增量

在 R^3 的 1-form
有 3 个基本形式 1-form
 $dx \quad dy \quad dz$

一般形式 1-form

$$\omega = P dx + Q dy + R dz$$

$$F = \begin{bmatrix} P(x,y,z) \\ Q(x,y,z) \\ R(x,y,z) \end{bmatrix}$$

$$\omega_F = P dx + Q dy + R dz$$

$$\omega_F(v) = P v_1 + Q v_2 + R v_3$$

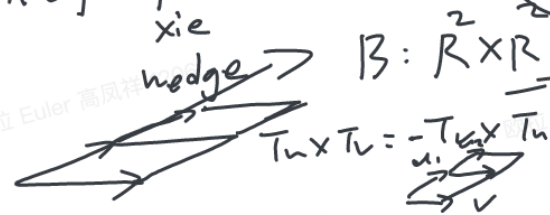
$$\omega_F = F \cdot v$$

$$\int_{\gamma(t)} \omega_F = \int F(\gamma'(t)) dt$$

$$= \int F \cdot \gamma'(t) dt$$

向量线积分
环流、做功

$dx \wedge dy$ 楔积
wedge



$$dx \wedge dy = -dy \wedge dx$$



$$\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\vec{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\vec{u} = u_1 \vec{e}_1 + u_2 \vec{e}_2$$

$$\vec{v} = v_1 \vec{e}_1 + v_2 \vec{e}_2$$

$$1. \quad B(u, v) = -B(v, u)$$

$$2. \quad B(a\vec{u}_1 + b\vec{u}_2, \vec{v})$$

$$= aB(\vec{u}_1, \vec{v}) + bB(\vec{u}_2, \vec{v})$$

$$B(\vec{u}, a\vec{v}_1 + b\vec{v}_2)$$

$$=$$

$$B(\vec{u}, \vec{v})$$

$$= B(u_1 \vec{e}_1 + u_2 \vec{e}_2, v_1 \vec{e}_1 + v_2 \vec{e}_2)$$

$$= u_1 B(\vec{e}_1, v_1 \vec{e}_1 + v_2 \vec{e}_2) + u_2 B(\vec{e}_2, v_1 \vec{e}_1 + v_2 \vec{e}_2)$$

$$= u_1 v_1 B(\vec{e}_1, \vec{e}_1) + u_1 v_2 B(\vec{e}_1, \vec{e}_2)$$

$$+ u_2 v_1 B(\vec{e}_2, \vec{e}_1) + u_2 v_2 B(\vec{e}_2, \vec{e}_2)$$

$$= (u_1 v_2 - u_2 v_1) B(\vec{e}_1, \vec{e}_2)$$

$$= \det \begin{bmatrix} u_1 & u_2 \\ v_1 & v_2 \end{bmatrix}$$

$$B(\vec{u}, \vec{v}) = \det \begin{bmatrix} \dot{u}_1 & \dot{v}_1 \\ \dot{u}_2 & \dot{v}_2 \\ \dot{u}_3 & \dot{v}_3 \end{bmatrix}$$

$$dx \wedge dy(\vec{u}, \vec{v}) = \det \begin{bmatrix} \dot{u}_1 & \dot{v}_1 \\ \dot{u}_2 & \dot{v}_2 \end{bmatrix}$$

$$u = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \quad v = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

$$u \times v = \begin{bmatrix} dy \wedge dz \\ dz \wedge dx \\ dx \wedge dy \end{bmatrix}$$

$$dx \wedge dy(u, v) = \det \begin{bmatrix} u_1 & v_1 \\ u_2 & v_2 \end{bmatrix}$$

$$dy \wedge dz(u, v) = \det \begin{bmatrix} u_2 & v_2 \\ u_3 & v_3 \end{bmatrix}$$

$$dz \wedge dx(u, v) = \det \begin{bmatrix} u_1 & v_1 \\ u_3 & v_3 \end{bmatrix}$$

$$F = \begin{bmatrix} P(x, y, z) \\ Q(x, y, z) \\ R(x, y, z) \end{bmatrix}$$

$$\frac{W_F(T_u, T_v)}{F \cdot (T_u \times T_v)}$$

一般形式 2-form

$$\int_{\Phi(u, v)} \omega_F =$$

$$\omega = P dy \wedge dz + Q dz \wedge dx + R dx \wedge dy$$

$$= \begin{bmatrix} P & 1 & 1 \\ Q & T_u & T_v \\ R & 1 & 1 \end{bmatrix}$$